

MC8P MIDI Controller - Owner's Manual

.axs instruments

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Introduction

The MC8P MIDI Controller is a compact, TRS Type A MIDI device designed to enhance hands-on control of the OP-XY by Teenage Engineering. Featuring 8 fully assignable potentiometers, this controller provides immediate access to critical parameters—eliminating menu-diving and streamlining your live performance workflow.

Built around the Teensy 4.0 microcontroller, the MC8P is an open-source project, offering both reliability and customization. I developed this controller out of necessity, seeking a more fluid and intuitive way to manipulate the OP-XY in real time. Whether tweaking filters, modulating effects, or shaping envelopes, the MC8P puts essential controls at your fingertips.

This manual will guide you through setup, configuration, and operation—helping you unlock the full potential of your MC8P.

User Guide

1.0 PERFORMANCE Mode (Main Screen)

- Displays real-time MIDI control message (CC) values for each potentiometer.
- No assignments can be edited here—this is purely for live control.
- To enter ASSIGN Mode: Hold the ASSIGN button for 2 seconds. The screen will flash to confirm.
- There are no other button configurations yet - updates may come in the future.

1.1 TEMP Mode

TEMP Mode allows you to make temporary adjustments to multiple parameters at once, perfect for experimenting with sounds or building dramatic transitions—without permanently altering your MIDI controller’s active values.

How to use TEMP Mode:

- While in PERFORMANCE Mode (Main Screen), press and hold the ENTER button.
- With ENTER held, the screen will invert. You can freely adjust any of the 8 potentiometers and their MIDI CC values will be sent in real time.
- When you release the ENTER button:
 - All potentiometer values instantly return to the positions they were in before you pressed ENTER.
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 - The physical knobs will then “catch up” smoothly to match those saved positions as you continue to turn them.

Use case example:

You can temporarily swell filter cutoffs, resonance, and effects sends to create a rising build-up, then release ENTER to snap everything back to your original mix—all without ever losing your starting point.

Note:

TEMP Mode does not save any changes to your controller’s memory. It is purely for live, non-destructive parameter manipulation.

2.1 ASSIGN Mode Screen Layout (Configuration Screen)

In this mode, you can assign, edit, or remove MIDI CCs for each potentiometer.

Screen Layout:

- Top Right: A diagram of all 8 potentiometers. The currently selected pot is highlighted with a star(-ish) symbol.
- MIDI CC List: Shows all assigned control messages for the selected potentiometer.
 - Use the arrow indicator (→) to see which CC is being edited.
 - The underlined parameter shows whether you're adjusting the MIDI Channel or CC Number.
- Note: Turning pots in this mode won't send MIDI data—changes take effect only in PERFORMANCE Mode.

2.2 ASSIGN Mode Navigation (Configuration)

Switch Pots/CCs:

- Hold ASSIGN + press NEXT/PREV to jump between potentiometers.
- Hold ENTER + press NEXT/PREV to cycle through CCs on the current pot.

Edit Parameters:

- Press ASSIGN to toggle between editing MIDI Channel or CC Number.
- Press NEXT/PREV to increment/decrement the selected value.

Add a New CC:

- Hold ENTER + NEXT (2 sec) to create a blank CC (default: Ch 1, CC1).
- Limit: 10 CCs per pot (to avoid visual clutter).

Delete a CC:

- Hold ENTER + PREV (2 sec) to remove the selected CC.

Save & Exit:

- Hold ASSIGN + ENTER (2 sec) to save to EEPROM and return to PERFORMANCE Mode.

Parameter Reset:

- Hold NEXT + PREV (5 sec) to restore defaults (*Ch 1–8, CC7 [OP-XY Track Volume]*).

A Note of Thanks

Thank you for purchasing the MC8P MIDI Controller and supporting this passion project of mine! As someone who loves making music and tinkering with hardware, creating this controller has been a deeply rewarding experience—and I'm incredibly grateful that you've chosen to be part of its journey.

The MC8P was born out of my own need for a more tactile, performance-friendly way to control the OP-XY, and I hope it brings the same fluidity and joy to your setup as it has to mine.

Your Feedback Matters

This is an evolving project, and I'd love to hear from you! If you have ideas for new features, improvements, or just want to share how you're using the MC8P, please reach out via:

Email: djaxiisuk@gmail.com

Instagram: [@axiismusicuk](https://www.instagram.com/axiismusicuk)

GitHub: [MC8P Midi Controller](#)

Your input helps shape future updates, so don't hesitate to get in touch.

Most of all—have fun! I hope the MC8P becomes an indispensable part of your creative process.